



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 5

1. **Title:** Implement localStorage and sessionStorage to save user preferences like theme settings.

2. **Software/Tools Required:** VS Code, JavaScript,html.

3. Theory:

- 1) Web Storage allows websites to store data in the browser using key-value pairs.
- 2) localStorage stores data persistently and remains available even after closing the browser.
- 3) sessionStorage stores data only for the current browser session.
- 4) setItem() is used to save the selected theme and storage type.
- 5) getItem() retrieves previously saved theme and storage information.
- 6) removeItem() removes stored preferences.
- 7) The program provides Light, Dark, and Blue themes using CSS classes.
- 8) classList.remove() and classList.add() are used to change the theme dynamically.
- 9) getElementById() accesses the required HTML elements.
- 10) innerText displays the current theme, storage type, saved value, and messages.
- 11) loadPreference() automatically retrieves saved preferences when the page loads.
- 12) savePreference() stores the selected theme in either localStorage or sessionStorage.
- 13) removePreference() clears the saved preferences and resets the theme to Light.
- 14) The program demonstrates web storage and dynamic DOM manipulation.



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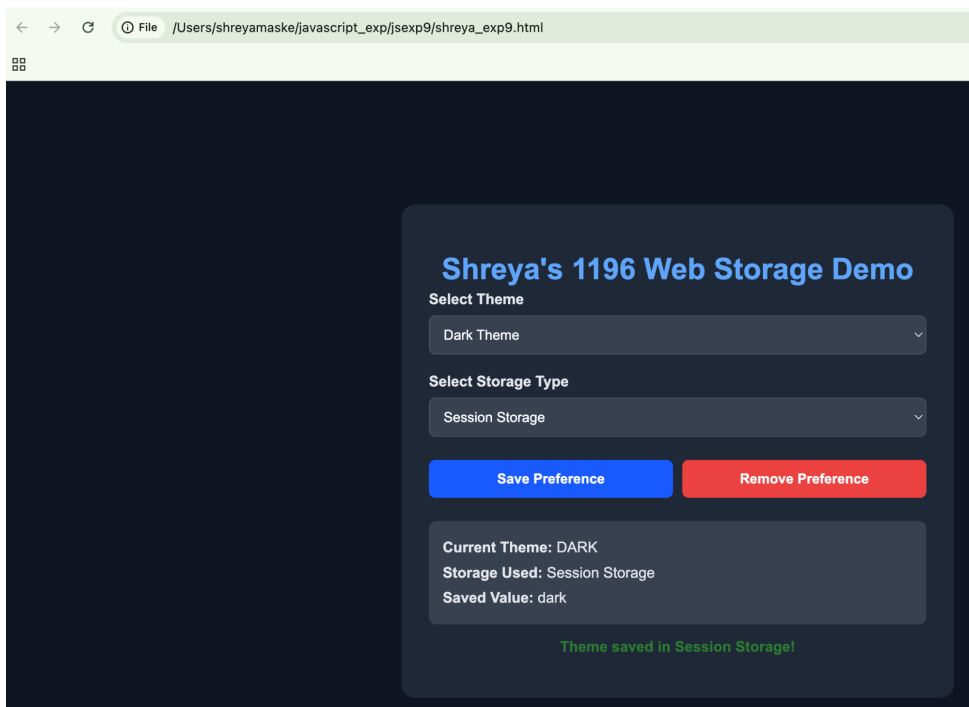
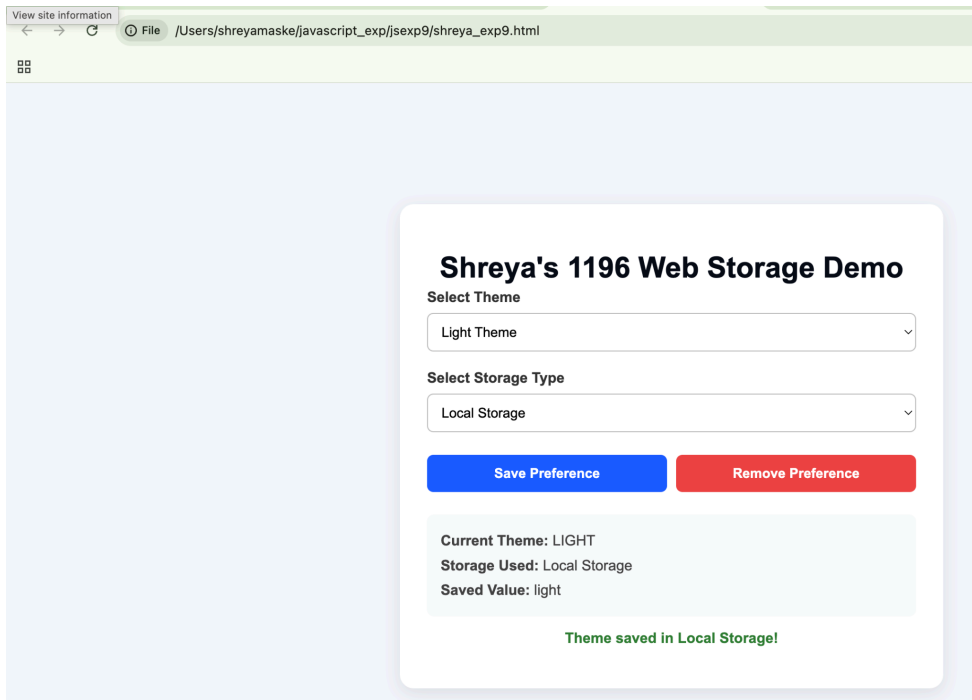
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4. Output :



5. Result/Conclusion:

Thus, local Storage and session Storage were successfully implemented to store and retrieve user theme preferences. The selected theme was applied dynamically, displayed on the webpage, automatically loaded when available, and removed successfully using JavaScript



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Case Study No.: 8

- Case Study Title:** Online Job Application Form, A company wants to create an online job application form that checks the entered information immediately and prevents submission when the details are invalid.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Job Application Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #dbeafe, #e0e7ff);
      margin: 0;
      padding: 25px;
    }
    .container {
      width: 650px;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 15px;
      box-shadow: 0 5px 20px rgba(0,0,0,0.15);
    }
    h2 {
      text-align: center;
      color: #3730a3;
    }
    h3 {
      color: #4f46e5;
      border-bottom: 2px solid #ddd;
      padding-bottom: 5px;
      margin-top: 25px;
    }
  </style>
</head>
</html>
```



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```
label {
    display: block;
    margin-top: 12px;
    font-weight: bold;
}

input, select, textarea {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    border: 1px solid #ccc;
    border-radius: 6px;
    box-sizing: border-box;
}

textarea {
    height: 70px;
    resize: none;
}

.radio {
    width: auto;
    margin-right: 5px;
}

.message {
    font-size: 13px;
    margin-top: 4px;
}

.error {
    color: red;
}

.success {
    color: green;
}

button {
    width: 100%;
    padding: 12px;
    margin-top: 20px;
    background-color: #4f46e5;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    cursor: pointer;
}
```



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```
}

button:hover {

    background-color: #3730a3;

}

#result {

    text-align: center;

    margin-top: 20px;

    font-weight: bold;

}

</style>
</head>
<body>
<div class="container">

    <h2>📁 Shreya's 1196 Online Job Application Form</h2>

    <form id="jobForm">

        <h3>Personal Information</h3>

        <!-- NAME -->

        <label>Full Name:</label>

        <input type="text" id="name" placeholder="Enter full name">

        <div id="nameMessage" class="message"></div>

        <!-- DOB -->

        <label>Date of Birth:</label>

        <input type="date" id="dob">

        <div id="dobMessage" class="message"></div>

        <!-- GENDER -->

        <label>Gender:</label>

        <input type="radio" class="radio" name="gender" value="Male">
Male

        <input type="radio" class="radio" name="gender" value="Female">
Female

        <input type="radio" class="radio" name="gender" value="Other">
Other

        <div id="genderMessage" class="message"></div>

        <!-- EMAIL -->

        <label>Email Address:</label>

        <input type="text" id="email" placeholder="Enter email">

        <div id="emailMessage" class="message"></div>

        <!-- MOBILE -->

        <label>Mobile Number:</label>

        <input type="text" id="mobile"

            placeholder="Enter 10-digit mobile number"

    </div>

</form>

</div>

</body>

</html>
```



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```
maxlength="10">

<div id="mobileMessage" class="message"></div>

<!-- ADDRESS -->
<label>Address:</label>
<textarea id="address" placeholder="Enter address"></textarea>
<div id="addressMessage" class="message"></div>

<!-- CITY -->
<label>City:</label>
<input type="text" id="city" placeholder="Enter city">
<div id="cityMessage" class="message"></div>

<!-- STATE -->
<label>State:</label>
<input type="text" id="state" placeholder="Enter state">
<div id="stateMessage" class="message"></div>

<!-- PIN -->
<label>PIN Code:</label>
<input type="text" id="pin"
      placeholder="Enter 6-digit PIN"
      maxlength="6">
<div id="pinMessage" class="message"></div>

<!-- QUALIFICATION -->
<label>Educational Qualification:</label>
<select id="qualification">
  <option value="">Select qualification</option>
  <option>B.Tech</option>
  <option>B.E.</option>
  <option>BCA</option>
  <option>MCA</option>
  <option>M.Tech</option>
  <option>Other</option>
</select>
<div id="qualificationMessage" class="message"></div>

<!-- PERCENTAGE -->
<label>Percentage:</label>
<input type="number" id="percentage"
      placeholder="Enter percentage">
<div id="percentageMessage" class="message"></div>

<!-- PASSING YEAR -->
<label>Passing Year:</label>
<input type="number" id="passingYear"
      placeholder="Enter passing year">
```



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```
<div id="yearMessage" class="message"></div>

<h3>Professional Information</h3>

<!-- POSITION -->
<label>Applying for Job Position:</label>
<input type="text" id="position"
        placeholder="Example: Software Developer">
<div id="positionMessage" class="message"></div>

<!-- EXPERIENCE -->
<label>Total Work Experience (Years):</label>
<input type="number" id="experience"
        placeholder="Enter experience">
<div id="experienceMessage" class="message"></div>

<!-- ORGANIZATION -->
<label>Current / Previous Organization:</label>
<input type="text" id="organization"
        placeholder="Enter organization">
<div id="organizationMessage" class="message"></div>

<!-- DESIGNATION -->
<label>Current Designation:</label>
<input type="text" id="designation"
        placeholder="Example: Intern">
<div id="designationMessage" class="message"></div>

<!-- SALARY -->
<label>Expected Salary:</label>
<input type="number" id="salary"
        placeholder="Enter expected salary">
<div id="salaryMessage" class="message"></div>

<!-- SKILLS -->
<label>Technical Skills:</label>
<textarea id="skills"
        placeholder="Example: Java, Python, SQL,
HTML"></textarea>
<div id="skillsMessage" class="message"></div>
<button type="submit">Submit Application</button>
</form>

<div id="result"></div>
</div>
<script>
    let form = document.getElementById("jobForm");
    // FUNCTION TO SHOW MESSAGE
    function showMessage(id, message, type) {
```



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```
document.getElementById(id).className = "message " + type;
document.getElementById(id).innerHTML = message;

// NAME
document.getElementById("name").addEventListener("input", function()
{
    if (this.value.trim() === "") {
        showMessage("nameMessage", "Name is required", "error");
    } else {
        showMessage("nameMessage", "✓ Valid name", "success");
    }
});

// DATE OF BIRTH
document.getElementById("dob").addEventListener("change", function()
{
    if (this.value === "") {
        showMessage("dobMessage", "Date of birth is required",
"error");
    } else {
        showMessage("dobMessage", "✓ Date of birth selected",
"success");
    }
});

// GENDER
document.querySelectorAll('input[name="gender"]').forEach(function(radi
o) {
    radio.addEventListener("change", function() {
        showMessage("genderMessage", "✓ Gender selected",
"success");
    });
});

// EMAIL
document.getElementById("email").addEventListener("input",
function() {
    let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (pattern.test(this.value)) {
        showMessage("emailMessage", "✓ Valid email", "success");
    } else {
        showMessage("emailMessage", "Invalid email address",
"error");
    }
});
```



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```
    }

    });

    // MOBILE
    document.getElementById("mobile").addEventListener("input",
function() {
    this.value = this.value.replace(/^[^0-9]/g, "");
    if (this.value.length === 10) {
        showMessage("mobileMessage", "✓ Valid mobile number",
"success");
    } else {
        showMessage("mobileMessage", "Invalid mobile number - enter
10 digits", "error");
    }
});

    // ADDRESS
    document.getElementById("address").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("addressMessage", "Address is required",
"error");
    } else {
        showMessage("addressMessage", "✓ Valid address",
"success");
    }
});

    // CITY
    document.getElementById("city").addEventListener("input", function()
{
    if (this.value.trim() === "") {
        showMessage("cityMessage", "City is required", "error");
    } else {
        showMessage("cityMessage", "✓ Valid city", "success");
    }
});

    // STATE
    document.getElementById("state").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("stateMessage", "State is required", "error");
    } else {
        showMessage("stateMessage", "✓ Valid state", "success");
    }
});
});
```



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```
}

});

// PIN
document.getElementById("pin").addEventListener("input", function()
{
    this.value = this.value.replace(/^[^0-9]/g, "");
    if (this.value.length === 6) {
        showMessage("pinMessage", "✓ Valid PIN code", "success");
    } else {
        showMessage("pinMessage", "Invalid PIN - enter 6 digits",
"error");
    }
});

// QUALIFICATION
document.getElementById("qualification").addEventListener("change",
function() {
    if (this.value === "") {
        showMessage("qualificationMessage",
            "Please select qualification", "error");
    } else {
        showMessage("qualificationMessage",
            "✓ Qualification selected", "success");
    }
});

// PERCENTAGE
document.getElementById("percentage").addEventListener("input",
function() {
    let value = Number(this.value);
    if (value >= 0 && value <= 100) {
        showMessage("percentageMessage",
            "✓ Valid percentage", "success");
    } else {
        showMessage("percentageMessage",
            "Invalid percentage - enter 0 to 100", "error");
    }
});

// PASSING YEAR
document.getElementById("passingYear").addEventListener("input",
function() {
    let year = Number(this.value);
    let currentYear = new Date().getFullYear();
```



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```
if (year >= 1950 && year <= currentYear) {
    showMessage("yearMessage",
        "✓ Valid passing year", "success");
} else {
    showMessage("yearMessage",
        "Invalid passing year", "error");
}
});
// JOB POSITION
document.getElementById("position").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("positionMessage",
            "Job position is required", "error");
    } else {
        showMessage("positionMessage",
            "✓ Valid job position", "success");
    }
});
// EXPERIENCE
document.getElementById("experience").addEventListener("input",
function() {
    let value = Number(this.value);

    if (value >= 0) {
        showMessage("experienceMessage",
            "✓ Valid experience", "success");
    } else {
        showMessage("experienceMessage",
            "Experience cannot be negative", "error");
    }
});
// ORGANIZATION
document.getElementById("organization").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("organizationMessage",
            "Organization is required", "error");
    } else {
        showMessage("organizationMessage",
```



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```
        "✓ Valid organization", "success");

    }

});

// DESIGNATION
document.getElementById("designation").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("designationMessage",
            "Designation is required", "error");
    } else {
        showMessage("designationMessage",
            "✓ Valid designation", "success");
    }
});

// SALARY
document.getElementById("salary").addEventListener("input",
function() {
    if (Number(this.value) > 0) {
        showMessage("salaryMessage",
            "✓ Valid expected salary", "success");
    } else {
        showMessage("salaryMessage",
            "Enter a valid salary", "error");
    }
});

// TECHNICAL SKILLS
document.getElementById("skills").addEventListener("input",
function() {

    if (this.value.trim() === "") {
        showMessage("skillsMessage",
            "Technical skills are required", "error");
    } else {
        showMessage("skillsMessage",
            "✓ Skills entered", "success");
    }
});
```



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```
// FORM SUBMISSION

form.addEventListener("submit", function(event) {

    event.preventDefault();

    let name = document.getElementById("name").value.trim();
    let dob = document.getElementById("dob").value;
    let gender =
document.querySelector('input[name="gender"]:checked');
    let email = document.getElementById("email").value;
    let mobile = document.getElementById("mobile").value;
    let address = document.getElementById("address").value.trim();
    let city = document.getElementById("city").value.trim();
    let state = document.getElementById("state").value.trim();
    let pin = document.getElementById("pin").value;
    let qualification =
document.getElementById("qualification").value;
    let percentage =
Number(document.getElementById("percentage").value);
    let passingYear =
Number(document.getElementById("passingYear").value);

    let position = document.getElementById("position").value.trim();
    let experience =
Number(document.getElementById("experience").value);
    let organization =
document.getElementById("organization").value.trim();
    let designation =
document.getElementById("designation").value.trim();
    let salary = Number(document.getElementById("salary").value);
    let skills = document.getElementById("skills").value.trim();

    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    let currentYear = new Date().getFullYear();
    if (
        name !== "" &&
        dob !== "" &&
        gender &&
        emailPattern.test(email) &&
        mobile.length === 10 &&
        address !== "" &&
        city !== "" &&
        state !== "" &&
```



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```
pin.length === 6 &&
qualification !== "" &&
percentage >= 0 &&
percentage <= 100 &&
passingYear >= 1950 &&
passingYear <= currentYear &&
position !== "" &&
experience >= 0 &&
organization !== "" &&
designation !== "" &&
salary > 0 &&
skills !== ""
) {
    document.getElementById("result").className = "success";

    document.getElementById("result").innerHTML =
        "✔ Application Submitted Successfully!";

    form.reset();

} else {

    document.getElementById("result").className = "error";
    document.getElementById("result").innerHTML =
        "✘ Please correct the invalid details before
submitting.";
    }
});
</script>
</body>
</html>
```

3. Output:



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File /Users/shreyamaske/javascript_exp/jsexp8/shreya_1196_case8.html



Shreya's 1196 Online Job Application Form

Personal Information

Full Name:

Shreya Maske

✓ Valid name

Date of Birth:

15 / 06 / 2006

✓ Date of birth selected

Gender:

☐ Male ☒ Female ☐ Other

✓ Gender selected

Email Address:

shreya@gmailcom

Invalid email address

Mobile Number:

7765211679

✓ Valid mobile number

Address:

412, Baker's Street

✓ Valid address

File /Users/shreyamaske/javascript_exp/jsexp8/shreya_1196_case8.html



City:

Nagpur

✓ Valid city

State:

Maharashtra

✓ Valid state

PIN Code:

44006

Invalid PIN - enter 6 digits

Educational Qualification:

B.Tech

✓ Qualification selected

Percentage:

88

✓ Valid percentage

Passing Year:

2028

Invalid passing year



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← → ↻ ⓘ File /Users/shreyamaske/javascript_exp/jsxp8/shreya_1196_case8.html

Professional Information

Applying for Job Position:

Software Developer

✓ Valid job position

Total Work Experience (Years):

-1

Experience cannot be negative

Current / Previous Organization:

xyz

✓ Valid organization

Current Designation:

Intern

✓ Valid designation

Expected Salary:

800000

✓ Valid expected salary

Technical Skills:

Java

✓ Skills entered

Submit Application

✗ Please correct the invalid details before submitting.

4. Result/Conclusion:

Thus, an online job application form was successfully created with **live validation** to check user details and prevent invalid form submission.